

$$\frac{d^2y}{dx^2} = \frac{dp}{dx} = p \frac{dp}{dy}$$

$$p^2 - yp \frac{dp}{dy} = 0.$$

$$y = c_2 e^{c_1 x}$$

U2

§1.

一般形式

$$\frac{d^ny}{dx^n} + p_1(x) \frac{d^{n-1}y}{dx^{n-1}} + \dots + p_n(x)y = f(x)$$

$f(x) \equiv 0$. 齐次线性微分方程

$$y^{(i)}|_{x=x_0} = y_0^{(i)}$$

$$y^{(i)}|_{x=x_0} = 0 \quad \text{唯一解 } y(x) \equiv 0.$$

$$L[y] \equiv \frac{d^ny}{dx^n} + p_1(x) \frac{d^{n-1}y}{dx^{n-1}} + \dots + p_n(x)y$$

$$2.1 \quad L[cy] = cL[y]$$

$$2.2 \quad L[y_1 + y_2] = L[y_1] + L[y_2]$$

$$2.3 \quad L\left[\sum_{i=1}^m c_i y_i\right] = \sum_{i=1}^m c_i L[y_i]$$

线性相关/无关,

$$a_1 y_1(x) + \dots + a_n y_n(x) \equiv 0 \quad \text{线性相关}$$

$$w(x) = \begin{vmatrix} y_1(x) & \cdots & y_m(x) \\ y_1'(x) & & y_m'(x) \\ \vdots & & \vdots \\ y_1^{(m-1)}(x) & & y_m^{(m-1)}(x) \end{vmatrix} = \phi(x)$$

为 $y_1(x) \dots y_m(x)$ 朗斯基行列式

齐次线性方程组不同解线性组合仍为解.

n 个线性无关解.

$y_1(x) \dots y_n(x)$ 为齐次方程 n 个解.

C_i 为常数. —————> 基本解组

$$y(x) = \sum_{i=1}^n C_i y_i(x) \text{ 为通解,}$$

$$y'' + p(x)y' + q(x)y = 0.$$

$$y(x) = C_1 y_1(x) + C_2 y_2(x)$$

非齐次.

① y_1^*, y_2^* 为非齐次解.

$\Rightarrow$ 则 $y_1^* - y_2^*$ 为齐次解.

② y^* 与 y 为非齐次解及对应解.

$y^* + y$ 也为解.

设 y^* : 非齐次解.

Y : 齐次方程通解.

$$y = Y + y^*.$$

$$L[y] = f_1(x) \quad L[y] = f_2(x)$$

$$L[y] = f_1(x) + f_2(x).$$

$$y = u(x) + i v(x)$$

$$L[y] = u(x) + i v(x)$$

§2 常系数线性微分方程.

一. $= P_{II}$ 对于 $\frac{d^2 y}{dx^2} + p \frac{dy}{dx} + qy = 0.$

先求线性无关解 y_1, y_2

$$s.t. \ y = C_1 y_1 + C_2 y_2.$$

$$y = e^{\lambda x}$$

$$\lambda^2 e^{\lambda x} + p \lambda e^{\lambda x} + q e^{\lambda x} = 0.$$

$$\lambda^2 + p\lambda + q = 0 \quad \text{特征方程} \quad \lambda: \text{特征根.}$$

$$y_1 = e^{\lambda_1 x} \quad y_2 = e^{\lambda_2 x} \quad \text{两个解.}$$

① $\lambda_1 \neq \lambda_2.$

$$\text{通解: } y = C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x}$$

② $\lambda_1 = \lambda_2.$

$$p^2 - 4q = 0. \quad \lambda_1 = \lambda_2 = -\frac{p}{2}$$

$$y_1 = e^{\lambda_1 x}$$

$$\text{取 } y = u e^{\lambda_1 x}$$

$$\frac{dy}{dx} = (u' + \lambda_1 u) e^{\lambda_1 x}$$

$$\frac{d^2 y}{dx^2} = (u'' + 2\lambda_1 u' + \lambda_1^2 u) e^{\lambda_1 x}$$

$$u'' + (2\lambda_1 p)u' + (\lambda_1^2 + p\lambda_1 + q)u = 0.$$

$$\lambda_1 = \text{重根}$$

$$\Rightarrow \lambda_1^2 + p\lambda_1 + q = 0 \quad 2\lambda_1 p = 0.$$

$$u'' = 0.$$

$$\text{取 } u = x.$$

$$\text{通解: } y = (c_1 + c_2 x) e^{\lambda_1 x}$$

$$(3) \lambda_1 = \alpha + i\beta \quad \lambda_2 = \alpha - i\beta.$$

$$y_1 = e^{(\alpha + i\beta)x} \quad y_2 = e^{(\alpha - i\beta)x}$$

$$e^{(\alpha \pm i\beta)x} = e^{\alpha x} (\cos \beta x \pm i \sin \beta x)$$

$$\frac{1}{2} (y_1 + y_2) = e^{\alpha x} \cos \beta x$$

$$\frac{1}{2i} (y_1 - y_2) = e^{\alpha x} \sin \beta x$$

两个解.

$$\text{通解: } y = c_1 e^{\alpha x} \cos \beta x + c_2 e^{\alpha x} \sin \beta x \\ = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$$

$$\text{或} = e^{\alpha x} \sqrt{c_1^2 + c_2^2} \sin(\beta x + \varphi) \\ = A e^{\alpha x} \sin(\beta x + \varphi)$$

$$\text{其中 } A = \sqrt{c_1^2 + c_2^2} \\ \sin \varphi = \frac{c_2}{\sqrt{c_1^2 + c_2^2}} \\ \cos \varphi = \frac{c_1}{\sqrt{c_1^2 + c_2^2}}$$

特征方程 $\lambda^2 + p\lambda + q = 0$ 的根	微分方程 $\frac{d^2 y}{dx^2} + p \frac{dy}{dx} + qy = 0$ 的通解
有不相等的实根 $\lambda_1 \neq \lambda_2$	$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$
有相等的实根 $\lambda_1 = \lambda_2$	$y = c_1 e^{\lambda_1 x} + c_2 x e^{\lambda_2 x}$
有共轭复数根 $\lambda_1 = \alpha + i\beta$ $\lambda_2 = \alpha - i\beta$	$y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$ 或 $y = A e^{\alpha x} \sin(\beta x + \varphi)$

$$\text{eg 1. } \frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} - 3y = 0.$$

$$\lambda^2 - 2\lambda - 3 = 0$$

$$\lambda_1 = 3 \quad \lambda_2 = -1$$

$$y = c_1 e^{3x} + c_2 e^{-x}$$

$$2. \frac{d^2 s}{dt^2} - 4 \frac{ds}{dt} + 4s = 0.$$

$$\lambda^2 - 4\lambda + 4 = 0 \quad \lambda_1 = \lambda_2 = 2$$

$$(C_1 + C_2 t) e^{2t}.$$

$$3. \frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + 13y = 0.$$

$$\lambda^2 + 4\lambda + 13 = 0.$$

$$\lambda = -2 \pm 3i$$

$$y = e^{-2x} (C_1 \cos 3x + C_2 \sin 3x)$$

$$\text{或 } y = A e^{-2x} \sin(3x + \varphi)$$

二. n 阶.

$$L[y] \equiv y^{(n)} + p_1 y^{(n-1)} + \dots + p_n y = 0$$

$$D(\lambda) = \lambda^n + p_1 \lambda^{n-1} + \dots + p_n = 0.$$

$$\text{通解: } y = \sum_{i=1}^s \sum_{j=1}^{n_i} c_{ij} x^{j-1} e^{\lambda_i x}$$

其中特征根 $\lambda_1, \dots, \lambda_s$.

重数 $n_1, \dots, n_s, \sum n_i = n$

$$\lambda_s \downarrow \begin{matrix} e^{\lambda_1 x}, x e^{\lambda_1 x}, \dots, x^{n_1-1} e^{\lambda_1 x} \\ \dots \\ e^{\lambda_s x}, \dots, x^{n_s-1} e^{\lambda_s x} \end{matrix}$$

$$e^{\alpha x} \cos \beta x, x e^{\alpha x} \cos \beta x, \dots, x^{n_1-1} e^{\alpha x} \cos \beta x$$

$$e^{\alpha x} \sin \beta x, \dots, x^{n_1-1} e^{\alpha x} \sin \beta x$$

特征方程(2.21) 的根	微分方程(2.20) 通解中对应的项
① 单重实根 λ	对应一项 $ce^{\lambda x}$
② k 重实根 λ	对应 k 项 $(c_1 + c_2 x + \dots + c_k x^{k-1}) e^{\lambda x}$
③ 单重复数根 $\lambda_{1,2} = \alpha \pm \beta i, \beta > 0$	对应两项 $e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$
④ k 重复数根 $\lambda_{1,2} = \alpha \pm \beta i, \beta > 0$	对应 $2k$ 项 $e^{\alpha x} [(a_1 + a_2 x + \dots + a_k x^{k-1}) \cos \beta x + (b_1 + b_2 x + \dots + b_k x^{k-1}) \sin \beta x]$

eg1: $\frac{d^3 y}{dx^3} - 3 \frac{d^2 y}{dx^2} + 4y = 0$

$$\lambda^3 - 3\lambda^2 + 4 = (\lambda + 1)(\lambda - 2)^2 = 0.$$

$$y = c_1 e^{-x} + c_2 e^{2x} + c_3 x e^{2x}$$

eg2: $\frac{d^4 y}{dx^4} + k^4 y = 0.$

$$\lambda^4 + k^4 = 0. \Rightarrow (\lambda^2 + k^2 + \sqrt{2} \lambda k)(\lambda^2 + k^2 - \sqrt{2} \lambda k) = 0$$

$$+ \sqrt{2} k \lambda + k^2$$

$$\lambda = -\frac{\sqrt{2}}{2}k \pm i\frac{\sqrt{2}}{2}k \quad \lambda = \frac{\sqrt{2}}{2}k \pm i\frac{\sqrt{2}}{2}k$$

$$\text{三} \quad y'' + py' + qy = f(x)$$

$$\text{齐次} \quad y'' + py' + qy = 0$$

$$(-) f(x) = P_m(x) e^{\alpha x}$$

$$\frac{dy}{dx} + p \frac{dy}{dx} + qy = P_m(x) e^{\alpha x}$$

$$\text{取 } y^* = Q(x) e^{\alpha x}$$

$\times q$

$Q(x)$ 多项式

$$\frac{dy^*}{dx} = \frac{dQ}{dx} e^{\alpha x} + \alpha Q e^{\alpha x} \quad \times p$$

$$\frac{d^2 y^*}{dx^2} = \frac{d^2 Q}{dx^2} e^{\alpha x} + 2\alpha \frac{dQ}{dx} e^{\alpha x} + \alpha^2 Q e^{\alpha x}$$

代回.

$$\lambda^2 + (2\alpha + p)\lambda + (\alpha^2 + p\alpha + q) = P_m(x)$$

$$\frac{d^2 Q}{dx^2} + (2\alpha + p) \frac{dQ}{dx} + (\alpha^2 + p\alpha + q) Q = P_m(x)$$

$\uparrow$
m次

$$\textcircled{1} \alpha^2 + p\alpha + q \neq 0 \quad \alpha \text{ 不为根}$$

$$\text{令 } Q(x) = R_m(x) \quad \leftarrow m \text{ 次}$$

$$y^* = R_m(x) e^{\alpha x}$$

$$\textcircled{2} \alpha^2 + p\alpha + q = 0 \quad 2\alpha + p \neq 0$$

α 单根.

$$\boxed{\lambda^2 + p\lambda + q = 0}$$

特征方程

$$\text{令 } Q(x) = x R_m(x)$$

$$y^* = x R_m(x) e^{\alpha x}$$

$$\Delta = p^2 - 4q$$

$$\gamma^2$$

$$(3) \lambda^2 + p\lambda + q = 0. \quad 2\lambda + p = 0$$

$$\lambda = \text{重根}$$

$$\frac{d^2 Q}{dx^2} = p_m(x)$$

$$y^* = x^2 R_m(x) e^{\alpha x}$$

$$\text{总结为 } y^* = x^k R_m(x) e^{\alpha x}$$

$$k = \begin{cases} 0, & \lambda \text{ 不为特征根} \\ 1, & \lambda \text{ 单根} \\ 2, & \lambda \text{ 双根} \end{cases}$$

$$\text{eg. } \frac{d^2 y}{dx^2} + y = (x-2)e^{3x}$$

$$\lambda^2 + 1 \Rightarrow \lambda = \pm i$$

$$\frac{d^2 y}{dx^2} + y = 0 \quad : \quad y = C_1 \cos x + C_2 \sin x$$

$$\text{设 } y^* = (Ax+B)e^{3x}$$

$$(6+0)A + (3^2+1)(Ax+B) = x-2$$

$$2\lambda + p \quad \lambda^2 + p\lambda + q$$

$$A = \frac{1}{10} \quad B = -\frac{13}{50}$$

$$y^* = \left(\frac{1}{10}x - \frac{13}{50}\right)e^{3x}$$

$$y = Y + y^* = C_1 \cos x + C_2 \sin x + \left(\frac{1}{10}x - \frac{13}{50}\right)e^{3x}$$

eg. $y'' + py' + qy = 0$.

$$(C_1 + C_2 x)e^x$$

$$\lambda = 1$$

$$y'' - 2y' + y = x$$

$$y^* = Ax + B \quad \text{代入}.$$

$$A = 1 \quad B = 2$$

$$\therefore y = (C_1 + C_2 x)e^x + x + 2$$

$$(=) f(x) = P_m(x) e^{ax} \cos bx / \sin bx$$

$$P_m(x) e^{(a+ib)x} \text{ 实部}$$

$$\text{考虑 } y'' + py' + qy = P_n(x) e^{(a+ib)x}$$

$$\text{eg } y'' - y = e^x \sin 2x$$

$$y'' - y = 0 \quad \lambda = \pm 1$$

$$Y = C_1 e^x + C_2 e^{-x}$$

$\lambda = 1+2i$ 不为解

$$\text{设 } y = A e^{(1+2i)x}$$

$$A(1+2i)^2 - A = 1$$

$$A = -\frac{1}{8}(i+1)$$

$$y = -\frac{1}{8}(i+1)e^{(1+2i)x}$$

$$= -\frac{1}{8}(i+1)e^x (\cos 2x + i \sin 2x)$$

$$y^* = -\frac{1}{8}e^x (\cos 2x + i \sin 2x)$$

$$y = Y + y^* = C_1 e^x + C_2 e^{-x}$$

$$- \frac{1}{8}e^x (\cos 2x + i \sin 2x)$$

$$\text{又 } y'' + py' + qy = f(x)$$

$$y^* = x^k (R_n(x)e^{\alpha x} \cos bx + S_n(x)e^{\alpha x} \sin bx)$$

$$其中 h = \max\{m, L\}$$

$$k = \begin{cases} 0 & , a \pm bi \text{ 不为特征根} \\ 1 & , a \pm bi \text{ 为单特征根} \end{cases}$$

$$y'' + 4y = 2\cos 2x$$

$$\lambda^2 + 4$$

$$C_1 \cos 2x + C_2 \sin 2x \quad 0 \pm 2i$$

$$y^* = x(A \cos 2x + B \sin 2x)$$

$$\frac{d^2x}{dy^2} + (y + \sin^2 x) \left(\frac{dx}{dy} \right)^2.$$

$$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$$

$$\frac{d^2x}{dy^2} = - \frac{\frac{d^2y}{dx^2}}{\left(\frac{dy}{dx} \right)^3}$$

$$\frac{d^2y}{dx^2} - y = \sin^2 x = \frac{1}{2} - \frac{1}{2} \cos 2x$$

$$y = C_1 e^x + C_2 e^{-x}$$

$$\frac{1}{2} y_1'' = -\frac{1}{2}$$

$$-\frac{1}{2} \cos 2x; \quad y_2'' = -\frac{1}{2} + \frac{1}{10} \cos 2x$$

$$y = C_1 e^x + C_2 e^{-x} - \frac{1}{2} + \frac{1}{10} \cos 2x$$

推广

$$L[y] = f(x)$$

① 设 $f(x) = P_m(x) e^{\alpha x}$

$$k = \begin{cases} 3, & \alpha \text{ 为三重根} \\ \vdots \end{cases}$$

② 设 $f(x) = P_m(x) e^{\alpha x} \cosh x + Q_t(x) e^{\alpha x} \sinh x$

$$k = \begin{cases} 2, & a \pm bi = \text{重根} \\ \vdots \end{cases}$$

eg. $y''' + 3y'' + 3y' + y = (x-5)e^{-x}$

$$\lambda^3 + 3\lambda^2 + 3\lambda + 1 = 0$$

$$(\lambda+1)^3 = 0$$

$$\lambda = -1$$

$$\alpha = -1, \text{ 三重根}$$

$$Y = (C_1 + C_2 x + C_3 x^2) e^{-x}$$

$$\text{设 } y^* = x^3(Ax+B)e^{-x} \\ = (Ax^4+Bx^3)e^{-x}$$

$$\text{求 } y^{*'}, y^{*''}, y^{*'''} \quad \text{求}$$

$$\begin{cases} (24A-18B)x + 6B + 18Bx = x-5 \end{cases}$$

$$A = \frac{1}{24} \quad B = -\frac{5}{6}$$

$$y = Y + y^* = (C_1 + C_2x + C_3x^2 - \frac{5}{6}x^3 + \frac{1}{24}x^4)e^{-x}$$

$$\text{eg. } f(x) = e^{-x} + \frac{1}{2} \int_0^x (x-t)^2 f(t) dt$$

$$\parallel \\ \frac{1}{2} \int_0^x (x^2 - 2xt + t^2) f(t) dt$$

$$= \frac{1}{2} x^2 \int_0^x f(t) dt - x \int_0^x t f(t) dt + \frac{1}{2} \int_0^x t^2 f(t) dt$$

$$f'(x) = -e^{-x} + x \int_0^x f(t) dt + \frac{x^2}{2} f(x) - x^2 f(x) - \int_0^x t f(t) dt + \frac{1}{2} x^2 f(x)$$

$$= -e^{-x} + x \int_0^x f(t) dt - \int_0^x t f(t) dt$$

$$f''(x) = e^{-x} + \int_0^x f(t) dt$$

$$f'''(x) = -e^{-x} + f(x)$$

$$f(0) = 1 \quad f'(0) = -1 \quad f''(0) = 1$$

$$f^{(4)}(x) - f(x) = -e^{-x}$$

$$\lambda^3 - 1 = 0$$

$$(\lambda - 1)(\lambda^2 + \lambda + 1) = 0$$

$$\lambda_1 = 1 \quad \lambda_{2,3} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$$

$$f(x) = C_1 e^x + e^{-\frac{1}{2}x} \left(C_2 \cos \frac{\sqrt{3}}{2}x + C_3 \sin \frac{\sqrt{3}}{2}x \right)$$

$$\text{取 } f^*(x) = A e^{-x}$$

P 95

§4 欧拉方程

$$a_0 x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} x \frac{dy}{dx} + a_n y = f(x)$$

(-) 变量代换 $x = e^t \quad t = \ln x$

$$\text{eg. } a_0 x^2 \frac{d^2 y}{dx^2} + a_1 x \frac{dy}{dx} + a_2 y = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{1}{x} \cdot \frac{dy}{dt} \quad \begin{matrix} y = x^2 \\ t = \ln x \end{matrix}$$

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{1}{x} \cdot \frac{dy}{dt} \right)$$

$$= \frac{1}{x^2} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) \quad \frac{1}{x^2} (y'' - y')$$

$$\therefore a_0 \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) + a_1 \frac{dy}{dt} + a_2 y = f(e^t)$$

$$\text{即 } a_0 \frac{d^2 y}{dt^2} - (a_1 - a_0) \frac{dy}{dt} + a_2 y = f(e^t)$$

(二) 降阶

$$\frac{d^2 y}{dx^2} + p(x) \frac{dy}{dx} + q(x)y = 0.$$

若有非0解 y_1

引入 u . 取 $y = y_1 u$

$$y' = y_1 u' + y_1' u$$

$$y'' = y_1 u'' + 2y_1' u' + y_1 u''$$

$$\text{即 } y_1 u'' + [2y_1' + p(x)y_1]u' + \underbrace{[y_1'' + p(x)y_1' + q(x)y_1]}_{=0} u = 0.$$

$$\frac{1}{2} u' = z$$

$$\text{即 } y_1 \frac{dz}{dx} + [2y_1' + p(x)y_1]z = 0$$

$$z = \frac{C_2}{y_1^2} e^{-\int p(x) dx}$$

$$y = y_1 \left[C_1 + C_2 \int \frac{1}{y_1^2} e^{-\int p(x) dx} dx \right]$$

Liouville 公式

(三) 特殊

$$2p'(x) + p^2(x) - 4q(x) = 0$$

$$\frac{d^2 y}{dx^2} + p(x) \frac{dy}{dx} + q(x)y = 0.$$

$$y = uv \quad v = v(x)$$

$$u''v + u'(2v' + pv) + u(v'' + pv' + qu)$$

$$v: 2v' + pv = 0.$$

$$v = e^{-\int \frac{p}{2} dx}$$

$$u'' + pv' + qu = -\frac{1}{4}(2p' + p^2 - 4q)e^{-\int \frac{p}{2} dx}$$

$$= -\frac{a}{4}e^{-\int \frac{p}{2} dx}$$

$$u'' - \frac{a}{4}u = 0.$$

eg. $4 \frac{d^2 y}{dx^2} + 4x \frac{dy}{dx} + (x^2 + 1)y = 0.$

$$\Rightarrow \frac{d^2 u}{dx^2} + \lambda u = 0.$$

$$\Rightarrow \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + \frac{1}{4}(x^2 + 1)y = 0.$$

$$p(x) = x \quad q(x) = \frac{1}{4}(x^2 + 1),$$

$$2p'(x) + p^2(x) - 4q(x) = 1$$

$$a = 1$$

$$\Rightarrow y = u e^{-\int \frac{p}{2} dx} = u e^{-\frac{x^2}{4}}$$

$$u'' - \frac{1}{4}u = 0$$

$$u = c_1 e^{\frac{x}{2}} + c_2 e^{-\frac{x}{2}}$$

$$y = e^{-\frac{x^2}{4}} (c_1 e^{\frac{x}{2}} + c_2 e^{-\frac{x}{2}})$$

二、变动任意常数

$$y'' + p_1(x)y' + p_2(x)y = f(x)$$

通解 $y = c_1 y_1(x) + c_2 y_2(x)$.

$$y = u_1 y_1(x) + u_2 y_2(x)$$

①

$$\{ \bigcap u_1, u_2$$

$$y' = [u_1 y_1'(x) + u_2 y_2'(x)] + [u_1' y_1(x) + u_2' y_2(x)]$$

使 $u_1' y_1(x) + u_2' y_2(x) = 0$

$$R)y' = u_1 y_1'(x) + u_2 y_2'(x).$$

$$y'' = [u_1 y_1''(x) + u_2 y_2''(x)] + [u_1' y_1'(x) + u_2' y_2'(x)]$$

$$\begin{cases} u_1' y_1(x) + u_2' y_2(x) = 0 \\ u_1' y_1'(x) + u_2' y_2'(x) = f(x) \end{cases}$$

$$u_1' = - \frac{y_2(x)}{u_1(x)} f(x).$$

$$u_2' = \frac{y_1(x)}{w(x)} f(x).$$

$$u_1 = \int \frac{y_2(x)}{w(x)} f(x) dx$$

$\Rightarrow$

$$u_2 = \int \frac{y_1(x)}{w(x)} f(x) dx.$$

代入 ①

$$y^* = -y_1(x) \int \frac{y_2(x)}{w(x)} f(x) dx \\ + y_2(x) \int \frac{y_1(x)}{w(x)} f(x) dx$$

$$y = y_1(x) \left(C_1 - \int \frac{y_2(x)}{w(x)} f(x) dx \right)$$

$$+ y_2(x) \left(C_2 + \int \frac{y_1(x)}{w(x)} f(x) dx \right)$$

eg. 求 $y'' + a^2 y = f(x)$ 通解.

$$y = C_1 \cos ax + C_2 \sin ax$$

变力/生变/变力.

$$y = u_1 \cos ax + u_2 \sin ax$$

$$y' = -a u_1 \sin ax + a u_2 \cos ax \\ + u_1' \cos ax + u_2' \sin ax$$

$$\frac{1}{2} u_1' \cos ax + u_2' \sin ax = 0.$$

$$y' = -a u_1 \sin ax + a u_2 \cos ax$$

$$y'' = -a^2 u_1 \cos ax - a^2 u_2 \sin ax \\ - a u_1' \sin ax + a u_2' \cos ax$$

代入

$$- a u_1' \sin ax + a u_2' \cos ax = f(x)$$

$$u_1' \cos ax + u_2' \sin ax = 0.$$

$$u_1' = -\frac{1}{a} f(x) \sin ax$$

$$u_2' = \frac{1}{a} f(x) \cos ax$$

$$u_1 = -\frac{1}{a} \int_{x_0}^x f(\xi) \sin a\xi d\xi$$

$$u_2 = \frac{1}{a} \int_{x_0}^x f(\xi) \cos a\xi d\xi$$

$$y^* = \frac{1}{a} \int_{x_0}^x f(\xi) \sin a(x-\xi) d\xi$$

$$y = C_1 \cos ax + C_2 \sin ax + \frac{1}{a} \int_{x_0}^x f(\xi) \sin a(x-\xi) d\xi$$

eg. $y'' + py' + qy = 0$ 通解 $(C_1 + C_2 x)e^x$

$$y_1(x) = e^x \quad y_2(x) = x e^x$$

$$w(x) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} = \begin{vmatrix} e^x & x e^x \\ e^x & (1+x)e^x \end{vmatrix} = e^{2x}$$

$$y(x) = e^x (C_1 - \int x e^x \cdot x e^{-2x} dx)$$

$$+xe^x(C_2 + \int e^x \cdot xe^{-2x} dx)$$

$$=(C_1 + C_2 x)e^x + x + 2$$

$$y(0)=2 \quad y'(0)=0.$$

↓

$$\text{特解 } y = -xe^x + x + 2.$$

三. 幂级数

$$y'' + y = 0$$

$$\text{设 } y = a_0 + a_1 x + \dots + a_n x^n + \dots$$

$$y' = a_1 + 2a_2 x + \dots + (n+1)a_{n+1}x^n$$

$$y'' = 2 \cdot 1a_2 + 3 \cdot 2a_3 x + \dots + (n+2)(n+1)a_{n+2}x^n$$

$$(2a_2 + a_0) + (6a_3 + a_1)x + (12a_4 + a_2)x^2 + \dots$$

$$[(n+2)(n+1)a_{n+2} + a_n]x^n = 0$$

$$y = a_0 \left(1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 \dots \right)$$

$$+ a_1 \left(x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 \dots \right)$$

Maclaurin 级数

$$a_0 \cos x + a_1 \sin x$$

$$y'' + p(x)y' + q(x)y = 0$$

$p(x), q(x)$ 在 $|x| < R$ 内部:

$$p(x) = \sum_{n=0}^{\infty} p_n x^n \quad q(x) = \sum_{n=0}^{\infty} q_n x^n.$$

$$y = \sum_{n=0}^{\infty} a_n x^n.$$

但 ν 阶 Bessel 方程.

$$x^2 y'' + x y' + (x^2 - \nu^2) y = 0 \quad (\nu \geq 0). \text{ 不满足,}$$

定理.

$$\text{设 } x^2 y'' + p(x) y' + q(x) y = 0.$$

$$p(0) = 0.$$

$$y = x^c \sum_{k=0}^{\infty} a_k x^k$$

$$R = \infty$$

$$y = x^c \sum_{k=0}^{\infty} x^k = \sum_{k=0}^{\infty} a_k x^{c+k}$$

$$\Rightarrow (c^2 - \nu^2) a_0 x^c + [(c+1)^2 - \nu^2] a_1 x^{c+1} + \sum_{k=2}^{\infty} \{[(c+k)^2 - \nu^2] a_k + a_{k-2}\} x^{c+k} = 0$$

$$\begin{cases} a_0(c^2 - \nu^2) = 0 \\ a_1[(c+1)^2 - \nu^2] = 0 \\ [(c+k)^2 - \nu^2] a_k + a_{k-2} = 0. \end{cases}$$

$$\textcircled{1} c = \nu \Rightarrow a_1 = 0.$$

$$J_{\nu}(x) = \sum_{m=0}^{\infty} \frac{(-1)^m x^{\nu+2m}}{2^{2m} m! (\Gamma(\nu+1)) (\nu+1) \cdots (\nu+m)}$$

ν 阶第一类 Bessel 函数